

Question No : 16 of 50

Let z be a complex valued function then
 $\sin^{-1} z =$ _____.

Answer



$$\log \left(iz + (1 - z^2)^{\frac{1}{2}} \right)$$

Not sure



$$\log \left(iz - (1 - z^2)^{\frac{1}{2}} \right)$$

04:20

Question No : 46 of 50

Prove that $u = y^3 - 3x^2y$ is harmonic.

10:41

Answer

← → | **B** *I* U | ✎ $\mathcal{I}_x$ | $\frac{1}{2}$ $\frac{3}{2}$ | $\frac{1}{4}$ $\frac{3}{4}$ | $\frac{1}{8}$ $\frac{3}{8}$ | $\frac{1}{16}$ $\frac{3}{16}$ | $\frac{1}{32}$ $\frac{3}{32}$ | $\frac{1}{64}$ $\frac{3}{64}$ | Styles | Format | Font

Given that :

$$u = y^3 - 3x^2y$$

AS we know that the HARMONIC is :

$$H + H = 0$$

Question No : 20 of 50

The derivative of a complex valued function $w(t) = u(t) + i v(t)$ is _____, where u, v are real valued functions of a real

Answer

05:03



$$w'(t) = u'(t) + v'(t)$$



$$w'(t) = u(t) + iv(t)$$

Evaluate $4^{\frac{1}{2}}$.

Answer

◀ ▶ **B** *I* U  *I*_x $\frac{1}{2}$ $\frac{3}{4}$ $\frac{5}{8}$ $\frac{7}{8}$ $\frac{9}{16}$ $\frac{11}{16}$ $\frac{13}{16}$ $\frac{15}{16}$ $\frac{17}{32}$ $\frac{19}{32}$ $\frac{21}{32}$ $\frac{23}{32}$ $\frac{25}{32}$ $\frac{27}{32}$ $\frac{29}{32}$ $\frac{31}{32}$ $\frac{33}{64}$ $\frac{35}{64}$ $\frac{37}{64}$ $\frac{39}{64}$ $\frac{41}{64}$ $\frac{43}{64}$ $\frac{45}{64}$ $\frac{47}{64}$ $\frac{49}{64}$ $\frac{51}{64}$ $\frac{53}{64}$ $\frac{55}{64}$ $\frac{57}{64}$ $\frac{59}{64}$ $\frac{61}{64}$ $\frac{63}{64}$ $\frac{65}{128}$ $\frac{67}{128}$ $\frac{69}{128}$ $\frac{71}{128}$ $\frac{73}{128}$ $\frac{75}{128}$ $\frac{77}{128}$ $\frac{79}{128}$ $\frac{81}{128}$ $\frac{83}{128}$ $\frac{85}{128}$ $\frac{87}{128}$ $\frac{89}{128}$ $\frac{91}{128}$ $\frac{93}{128}$ $\frac{95}{128}$ $\frac{97}{128}$ $\frac{99}{128}$ $\frac{101}{256}$ $\frac{103}{256}$ $\frac{105}{256}$ $\frac{107}{256}$ $\frac{109}{256}$ $\frac{111}{256}$ $\frac{113}{256}$ $\frac{115}{256}$ $\frac{117}{256}$ $\frac{119}{256}$ $\frac{121}{256}$ $\frac{123}{256}$ $\frac{125}{256}$ $\frac{127}{256}$ $\frac{129}{512}$ $\frac{131}{512}$ $\frac{133}{512}$ $\frac{135}{512}$ $\frac{137}{512}$ $\frac{139}{512}$ $\frac{141}{512}$ $\frac{143}{512}$ $\frac{145}{512}$ $\frac{147}{512}$ $\frac{149}{512}$ $\frac{151}{512}$ $\frac{153}{512}$ $\frac{155}{512}$ $\frac{157}{512}$ $\frac{159}{512}$ $\frac{161}{512}$ $\frac{163}{512}$ $\frac{165}{512}$ $\frac{167}{512}$ $\frac{169}{512}$ $\frac{171}{512}$ $\frac{173}{512}$ $\frac{175}{512}$ $\frac{177}{512}$ $\frac{179}{512}$ $\frac{181}{512}$ $\frac{183}{512}$ $\frac{185}{512}$ $\frac{187}{512}$ $\frac{189}{512}$ $\frac{191}{512}$ $\frac{193}{512}$ $\frac{195}{512}$ $\frac{197}{512}$ $\frac{199}{512}$ $\frac{201}{512}$ $\frac{203}{512}$ $\frac{205}{512}$ $\frac{207}{512}$ $\frac{209}{512}$ $\frac{211}{512}$ $\frac{213}{512}$ $\frac{215}{512}$ $\frac{217}{512}$ $\frac{219}{512}$ $\frac{221}{512}$ $\frac{223}{512}$ $\frac{225}{512}$ $\frac{227}{512}$ $\frac{229}{512}$ $\frac{231}{512}$ $\frac{233}{512}$ $\frac{235}{512}$ $\frac{237}{512}$ $\frac{239}{512}$ $\frac{241}{512}$ $\frac{243}{512}$ $\frac{245}{512}$ $\frac{247}{512}$ $\frac{249}{512}$ $\frac{251}{512}$ $\frac{253}{512}$ $\frac{255}{512}$ $\frac{257}{1024}$ $\frac{259}{1024}$ $\frac{261}{1024}$ $\frac{263}{1024}$ $\frac{265}{1024}$ $\frac{267}{1024}$ $\frac{269}{1024}$ $\frac{271}{1024}$ $\frac{273}{1024}$ $\frac{275}{1024}$ $\frac{277}{1024}$ $\frac{279}{1024}$ $\frac{281}{1024}$ $\frac{283}{1024}$ $\frac{285}{1024}$ $\frac{287}{1024}$ $\frac{289}{1024}$ $\frac{291}{1024}$ $\frac{293}{1024}$ $\frac{295}{1024}$ $\frac{297}{1024}$ $\frac{299}{1024}$ $\frac{301}{1024}$ $\frac{303}{1024}$ $\frac{305}{1024}$ $\frac{307}{1024}$ $\frac{309}{1024}$ $\frac{311}{1024}$ $\frac{313}{1024}$ $\frac{315}{1024}$ $\frac{317}{1024}$ $\frac{319}{1024}$ $\frac{321}{1024}$ $\frac{323}{1024}$ $\frac{325}{1024}$ $\frac{327}{1024}$ $\frac{329}{1024}$ $\frac{331}{1024}$ $\frac{333}{1024}$ $\frac{335}{1024}$ $\frac{337}{1024}$ $\frac{339}{1024}$ $\frac{341}{1024}$ $\frac{343}{1024}$ $\frac{345}{1024}$ $\frac{347}{1024}$ $\frac{349}{1024}$ $\frac{351}{1024}$ $\frac{353}{1024}$ $\frac{355}{1024}$ $\frac{357}{1024}$ $\frac{359}{1024}$ $\frac{361}{1024}$ $\frac{363}{1024}$ $\frac{365}{1024}$ $\frac{367}{1024}$ $\frac{369}{1024}$ $\frac{371}{1024}$ $\frac{373}{1024}$ $\frac{375}{1024}$ $\frac{377}{1024}$ $\frac{379}{1024}$ $\frac{381}{1024}$ $\frac{383}{1024}$ $\frac{385}{1024}$ $\frac{387}{1024}$ $\frac{389}{1024}$ $\frac{391}{1024}$ $\frac{393}{1024}$ $\frac{395}{1024}$ $\frac{397}{1024}$ $\frac{399}{1024}$ $\frac{401}{1024}$ $\frac{403}{1024}$ $\frac{405}{1024}$ $\frac{407}{1024}$ $\frac{409}{1024}$ $\frac{411}{1024}$ $\frac{413}{1024}$ $\frac{415}{1024}$ $\frac{417}{1024}$ $\frac{419}{1024}$ $\frac{421}{1024}$ $\frac{423}{1024}$ $\frac{425}{1024}$ $\frac{427}{1024}$ $\frac{429}{1024}$ $\frac{431}{1024}$ $\frac{433}{1024}$ $\frac{435}{1024}$ $\frac{437}{1024}$ $\frac{439}{1024}$ $\frac{441}{1024}$ $\frac{443}{1024}$ $\frac{445}{1024}$ $\frac{447}{1024}$ $\frac{449}{1024}$ $\frac{451}{1024}$ $\frac{453}{1024}$ $\frac{455}{1024}$ $\frac{457}{1024}$ $\frac{459}{1024}$ $\frac{461}{1024}$ $\frac{463}{1024}$ $\frac{465}{1024}$ $\frac{467}{1024}$ $\frac{469}{1024}$ $\frac{471}{1024}$ $\frac{473}{1024}$ $\frac{475}{1024}$ $\frac{477}{1024}$ $\frac{479}{1024}$ $\frac{481}{1024}$ $\frac{483}{1024}$ $\frac{485}{1024}$ $\frac{487}{1024}$ $\frac{489}{1024}$ $\frac{491}{1024}$ $\frac{493}{1024}$ $\frac{495}{1024}$ $\frac{497}{1024}$ $\frac{499}{1024}$ $\frac{501}{1024}$ $\frac{503}{1024}$ $\frac{505}{1024}$ $\frac{507}{1024}$ $\frac{509}{1024}$ $\frac{511}{1024}$ $\frac{513}{1024}$ $\frac{515}{1024}$ $\frac{517}{1024}$ $\frac{519}{1024}$ $\frac{521}{1024}$ $\frac{523}{1024}$ $\frac{525}{1024}$ $\frac{527}{1024}$ $\frac{529}{1024}$ $\frac{531$

Consider that :

$$z^c = 4^{\frac{1}{2}}$$

The Cauchy–Goursat theorem shows that _____.

Answer



$$\int_c f(z)dz = 2n\pi$$



$$\int_c f(z)dz = 0$$



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Question No : 29 of 50

Marks: 1 (Time

The Cauchy–Riemann equations on a pair of real-valued functions of two real variables $u(x, y)$ and $v(x, y)$ are _____.

Answer



$$U_x = V_y \text{ and } U_y = -V_x$$



$$U_x = V_x \text{ and } U_y = -V_y$$



$$U_y = V_y \text{ and } U_x = -V_x$$

07:10

Question No : 28 of 50

Marks: 1 (Time

Let $f(z) = 8z^2 - z$, then $f'(i)$ is _____.

06:54

Answer

☐ $8i-1$

☐ $16i$

☒ $16i-1$

☐ $8i$

Question No : 3 of 50

Marks: 1 (Tim

If a series of complex numbers converges, the n^{th} term converges to zero as n tends to _____.

01:39

Answer



infinity



finite numbers

The sequence $z_n = -2 + i \frac{1}{n^3}$ ($n = 1, 2, 3, \dots$) for $n \rightarrow \infty$ converges to _____.

07:47

Answer



-2



1

 i 

-1

In the complex valued function, $U(x, y)$ is harmonic in some domain if _____.

Answer



$$U_{xx} = U_{yy} = 0$$



$$U_{xx} + U_{yy} = 0$$



$$U_{xx} = U_{yy}$$

Question No : 35 of 50

The logarithmic function of $e^w = z$, where $z = re^{i\theta}$ and $w = u + iv$, then _____.

Answer



$$w = -\ln r + i\theta$$



$$w = \ln r + i(\theta + 2n\pi)$$

Question No : 11 of 50

In the complex valued function $f(z) = \frac{1}{r e^{i\theta}}$, the value of $u_r =$ _____.

Answer



$$\frac{1}{r}$$



$$-\frac{\cos \theta}{r^2}$$



$$\cos \theta$$

03:21

If $f(z) = z(2 - z)$, then $f(1 + i) =$ _____.

09:37

Answer



i



2



0



$-i$

Let x be a complex valued function then

$\sinh x =$ _____.

Answer



$$\frac{e^x + e^{-x}}{2}$$



$$\frac{e^x - e^{-x}}{2i}$$

04:27

Type here to search

Question No : 47 of 50

Compute the analytic region of $f(z) = (x - y)^2 + 2i(x + y)$.

10:59

Answer

← → | **B** *I* U | *I_x* | $\frac{1}{x}$ $\frac{1}{x^2}$ $\frac{1}{x^3}$ $\frac{1}{x^4}$ | $\frac{1}{x^5}$ $\frac{1}{x^6}$ $\frac{1}{x^7}$ $\frac{1}{x^8}$ | $\frac{1}{x^9}$ $\frac{1}{x^{10}}$ | Styles | Format | Font

Given fun :

$$f(z) = (x - y)^2 + 2i(x + y)$$

Question No : 21 of 50

The definite integral of a complex valued function $w(t) = u(t) + iv(t)$ over an interval $a \leq t \leq b$ is where u, v are real valued functions of a real variable t .

Answer



$$\int_a^b w(t) dt = \int_a^b u(t)v(t) dt$$



$$\int_a^b w(t) dt = \int_a^b u(t) dt + i \int_a^b v(t) dt$$

05.09

Using the Cauchy – Goursat theorem when the contour C is unit circle $\int_C ze^{-z} dz = \underline{\hspace{2cm}}$.

Answer

-1



1

 i 

0

In the complex valued function $z = x - i y$, the value of $U_x =$ _____.

Answer



-1



1



0



2

Question No : 33 of 50

_____ is the singularity of $f(z) = \frac{1}{z}$.

Answer

☐ $z = 2$

☒ $z = 1$

☐ $z = 3$

☐ $z = 0$

If U and V are harmonic functions then $f(Z) = U + iV$ is _____.

01:49

Answer

- ☐ analytic function.
- ☐ need not be analytic function.
- ☒ analytic function only at $Z=0$.
- ☐ none of the others.

Question No : 38 of 50

Harmonic conjugate of $u(x, y) = e^y \cos x$ is _____.

09:00

Answer



$$-e^y \sin x + c$$



$$e^x \cos y + c$$



$$e^x \sin y + c$$

Question No : 8 of 50

A real-valued function H of two variables x and y is said to be harmonic if H satisfies _____.

02:37

Answer



Cauchy Riemann equations



Laplacian equation

Give two examples of the entire function.

09:45

Answer

← → | **B** *I* U |  I_x | $\frac{1}{x}$ $\frac{1}{x^2}$ | $\frac{1}{x^3}$ $\frac{1}{x^4}$ | $\frac{1}{x^5}$ $\frac{1}{x^6}$ | $\frac{1}{x^7}$ $\frac{1}{x^8}$ | $\frac{1}{x^9}$ $\frac{1}{x^{10}}$ | $\frac{1}{x^{11}}$ $\frac{1}{x^{12}}$ | Styles ▼ | Format ▼ | F

1. Entire function is an analytic function which has their domain and range.
2. $\frac{d}{dx}\sin x = \cos x$

1. $\sin x$
2. $\cos x$
3. $\tan x$

Question No. 15 of 60

Let $f(z) = 3z^2 + 5z + 10$, then $f'(z)$ is _____.

Answer



$6z - 5$



$6z$



$6z + 5$



$5z$

03:39

$$f(z) = z - \bar{z}, \text{ where } z = x + iy.$$

10:31

$$f(z) = z - \bar{z}$$

For the function f and contour C use parametric representations for C to evaluate $\int_C f(z)dz$, if

a) $f(z) = z - 1$ and C is semi circle for $z = 1 + e^{i\theta}$ ($\pi \leq \theta \leq 2\pi$).

Answer

← → | **B** *I* U | *I_x* | $\frac{1}{x}$ $\frac{1}{x^2}$ $\frac{1}{x^3}$ $\frac{1}{x^4}$ | $\frac{1}{x^5}$ $\frac{1}{x^6}$ $\frac{1}{x^7}$ $\frac{1}{x^8}$ | $\frac{1}{x^9}$ $\frac{1}{x^{10}}$ | Styles | Format | Font

As we know that :

$$\int f(z)dz = \int_a^b f(z(t))z'(t)dt$$

The principal value of 1^i is _____.

Answer



-1

02:47



1



i



0

Question No : 14 of 50

In the trigonometric functions of complex numbers

$\sin x =$ _____.

Answer



$$\frac{e^{-ix} - e^{ix}}{2}$$



$$\frac{e^{ix} - e^{-ix}}{2i}$$

03:58

The circle $z = e^{i\theta}, 0 \leq \theta \leq 2\pi$ is centered at _____.

Answer

☒ (1,0)

☐ (0,0)

☐ (0,1)

☐ (1,1)

Let C denote a contour of length L , and suppose that a function $f(z)$ is piecewise continuous on C . If M is a non-negative constant such that $|f(z)| \leq M$ for all points z on C at which $f(z)$ is defined, then _____.

Answer



c



$$\left| \int_c f(z) dz \right| \leq ML$$

Question No : 34 of 50

For $|z_n - z| < \varepsilon$ whenever $n > n_0$, geometrically this mean that for sufficiently large values of n , the point z_n lie in any given _____ of z .

Answer



axis



ε neighborhood



Type here to search



5

Question No : 37 of 50

A function U is said to be harmonic if and only if _____.

Answer



$$U_x^2 + U_y^2 = 0$$



$$U_{xx} + U_{yy} = 0$$

08:50



Type here to search



Question No : 43 of 50

Let C denote the positively oriented boundary of the square whose sides lie along the lines

$x = \pm 2$ and $y = \pm 2$. Evaluate the integral $\int_C \frac{\cos z}{z(z^2 + 8)} dz$ by using Cauchy Integral formula.

Answer



B

I

U



***I*_x**

1₂

:₂



Styles



Question No : 1 of 50

Marks: 1 (Time 1 Min)

A function $f(z)$ is analytic function if _____ part $f(z)$ is analytic.

Answer

- ☐ (a) real
- ☐ (b) imaginary
- ☒ (c) both (a) and (b)

Let sequence $z_n = -2 + i \frac{(-1)^n}{n^2}$ ($n = 1, 2, 3, \dots$) and find $\lim_{n \rightarrow \infty} z_n$ converges or not and also find the value of ε .

10:18

Answer

← → **B** *I* U  T_x $\frac{1}{2} =$ $\bullet =$       Styles ▾ Format ▾ Font ▾

Given sequence is :

$$z_n = -2 + i \frac{(-1)^n}{n^2}$$

consider that:

The absolute convergence of a series of complex numbers implies the convergence of that _____.

02:00

Answer



sequence



series

Let $z \neq 0$ and c be a complex number, then $z^c =$ _____.

02:58

Answer

- ☐ $\exp[\log(z)]$
- ☐ $c \exp[\log(z)]$
- ☒ $\exp[c \log(z)]$

09:56

Answer

$$= i^i$$

In the trigonometric functions of complex numbers

$\cos x =$ _____.

Answer

☐

$$\frac{e^{ix} + e^{-ix}}{2}$$

04:08

☐

$$\frac{e^{-ix} - e^{ix}}{2}$$

Question No : 49 of 50

By using the Cauchy-Riemann equations (in polar form), show that $f'(z) = \frac{1}{2f(z)}$
 , where $f(z) = \sqrt{r} e^{\frac{-i\theta}{2}} ; (r > 0, \alpha < \theta < \alpha + 2\pi)$

11:15

Answer

Rich text editor toolbar: Undo, Redo, Bold, Italic, Underline, Text color, Background color, Bulleted list, Numbered list, Decrease indent, Increase indent, Table, Table of contents, Styles, Format, Font.

Question No : 19 of 50

The circle $z = e^{-i\theta}, 0 \leq \theta \leq 2\pi$ is oriented in _____ direction.

Answer

- ☐ downward
- ☒ counter-clockwise
- ☐ upward

Question No : 23 of 50

Let $f(z) = z$ be a continuous complex valued function, then $\int_0^{1+i} f(z) dz =$ _____.

Answer

☐ $1+i$

☐ 0

☒ i

not sure

☐ $1-i$

Question No : 27 of 50

Marks: 1 (Tim

By using the extension of Cauchy Integral formual $\int_c \frac{f(z)}{(z - z_0)^{n+1}} dz = \underline{\hspace{2cm}}$.

Answer

☐ $2\pi i f^{(n)}(z_0)$

☒ $f^{(n)}(z_0)$

$= 2\pi i / n! f^{(n)}(z_0)$

06:45

☐ $\frac{2\pi i}{n!} f^{(n)}(z_0)$

Any polynomial $a_0 + a_1z + a_2z^2 + \dots + a_nz^n$ ($a_n \neq 0$) of degree n ($n \geq 1$) has at least one zero. That is, there exists z_0 , such that $p(z_0) = 0$.

07:28

Answer

- ☐ no points
- ☐ n points
- ☐ 10 points
- ☒ one point

Question No : 2 of 50

Marks: 1 (Time 1 Min)



If $f(z) = x + ay + i(bx + cy)$ is analytic, then a, b, c equals to _____.

Answer



$c=1$ and $a=-b$



$a=1$ and $c=-b$



$b=1$ and $a=-c$



$a=b=c=1$

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Question No : 36 of 50

If z_1 and z_2 are two nonzero complex numbers then $\text{Log}(z_1 z_2) =$ _____.

08:36

Answer



$$\text{Log} z_1 + \text{Log} z_2$$



$$\text{Log} z_1 - \text{Log} z_2$$

A series converges to a number S if and only if the sequence of remainders tends to _____.

3 / 19



Answer



infinity



one



zero



Question No : 39 of 50

The points at which $f(z) = \frac{1}{z^2+1}$ is not analytic are _____.

Answer

09:09



i and $-i$



1 and i



-1 and $-i$